

Finite Element Methods

Theory, Geometry, and Computation

Aldebert Lecornu

École Polytechnique de Louvain
Université catholique de Louvain

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Chapter 1

Functional Spaces

In this chapter, we introduce Sobolev spaces, which can be viewed as a particular case of Hilbert spaces, for which the Lax–Milgram theorem can be stated. This illustrates the power of functional analysis: thinking of functions as points in infinite-dimensional vector spaces, where a solid—but linear—theory exists. Classical references for this material include [1, 4, 6, 2].

The aim of this chapter is to provide engineers with the vocabulary and basic tools necessary to read and understand scientific literature on Sobolev spaces in the context of the finite element method.

1.1 Introduction: the Dirichlet energy

We begin with a prototypical variational problem that naturally leads to the introduction of functional spaces.

The Dirichlet energy Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with sufficiently regular boundary (e.g., C^1, C^∞ or Lipschitz), and let $g : \partial\Omega \rightarrow \mathbb{R}$ be a given boundary datum. We consider the Dirichlet energy

$$\Pi(u) = \int_{\Omega} |\nabla u|^2 dx, \tag{1.1}$$

defined for functions $u : \Omega \rightarrow \mathbb{R}$ satisfying the boundary condition

$$u = g \quad \text{on } \partial\Omega.$$

The variational problem We seek to minimize $\Pi(u)$ among all admissible functions satisfying the prescribed boundary condition. Formally,

$$\inf \left\{ \int_{\Omega} |\nabla u|^2 dx : u = g \text{ on } \partial\Omega \right\}.$$

At this stage, the class of admissible functions is not yet specified precisely; this will motivate the introduction of Sobolev spaces. Recall that the $\inf$ (for infimum) refers to an eventually non attained minimization problem.

The direct method of the calculus of variations The direct method relies on three main ingredients [3]:

- the admissible set is nonempty,
- minimizing sequences exist,
- the underlying function space is complete (or compact), so that minimizing sequences admit limits

- The functional is continuous with respect to the notion of convergence minimizing sequences have.

The Dirichlet energy provides a simple and nontrivial setting in which these ideas can be illustrated explicitly. For instance $x \in \mathbb{R} \rightarrow |x|^2/2$ admits a minimum by this method.

Minimizing sequences Let $(u_k)_{k \in \mathbb{N}}$ be a minimizing sequence, i.e.

$$\Pi(u_k) \rightarrow \inf \Pi.$$

Minimizing sequences always exist by definition of the infimum. A key observation is that such sequences are automatically *Cauchy with respect to the gradient*. Indeed, using the quadratic structure of the energy, we compute, by the parallelogram identity,

$$\int_{\Omega} |\nabla u_m - \nabla u_n|^2 = 2 \int_{\Omega} |\nabla u_m|^2 + 2 \int_{\Omega} |\nabla u_n|^2 - \int_{\Omega} |\nabla u_m + \nabla u_n|^2. \quad (1.2)$$

The midpoint

$$v = \frac{u_m + u_n}{2}$$

is an admissible competitor, hence

$$\Pi(v) \geq \inf \Pi.$$

Expanding $\Pi(v)$ and comparing with $\Pi(u_m)$ and $\Pi(u_n)$ yields

$$\int_{\Omega} |\nabla u_m - \nabla u_n|^2 \leq 2(\Pi(u_m) + \Pi(u_n) - 2 \inf \Pi),$$

which tends to 0 as $m, n \rightarrow \infty$.

Motivation for functional spaces This argument shows that minimizing sequences are Cauchy in the natural norm associated with the gradient. To make this convergence meaningful and to ensure the existence of a limit, we must work in a complete function space, *i.e.* a function space where every Cauchy sequence converge. This naturally leads to the introduction of Sobolev spaces, which provide the appropriate framework for variational problems involving derivatives.

The necessity of enlarging the class of admissible functions can already be seen in simple one-dimensional examples. Let $\Omega = (-1, 1)$ and define

$$u_n(x) = \sqrt{\left(\frac{1}{n}\right)^2 + x^2}.$$

Then (u_n) is a sequence of $C^1(-1, 1)$ functions such that

$$\int_{-1}^1 |u'_m(x) - u'_n(x)|^2 dx \rightarrow 0 \quad \text{as } m, n \rightarrow \infty,$$

but the limit function

$$u(x) = |x|$$

does not belong to $C^1(-1, 1)$.

This highlights the contrast with finite-dimensional spaces such as $\mathbb{R}^N$, which are complete:

$$(x_n) \text{ Cauchy in } \mathbb{R}^N \implies x_n \text{ converges in } \mathbb{R}^N.$$

1.2 Weak derivatives

The example of the Dirichlet energy shows that minimizing sequences may converge to functions that are no longer classically differentiable. This motivates the introduction of a generalized notion of derivative, defined in a weak (or distributional) sense.

1.2.1 Definitions and general properties

Weak derivatives Let $\Omega \subset \mathbb{R}^n$ be an open set and let u be an integrable function on Ω , i.e. $u \in L^1(\Omega)$. We say that $v \in L^1(\Omega)$ is the *weak partial derivative* of u with respect to the variable x_i , and we write

$$v = \partial_i u,$$

if

$$\boxed{\int_{\Omega} u \partial_i \varphi \, dx = - \int_{\Omega} v \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\Omega).} \quad (1.3)$$

Test functions and the space $C_c^\infty(\Omega)$ We denote by $C_c^\infty(\Omega)$ the space of *test functions*, that is, the set of all functions

$$\varphi : \Omega \rightarrow \mathbb{R}$$

which are infinitely differentiable and have compact support in Ω . More precisely,

$$C_c^\infty(\Omega) := \{\varphi \in C^\infty(\Omega) ; \text{supp}(\varphi) \Subset \Omega\},$$

where $\text{supp}(\varphi)$ denotes the closure of the set

$$\{x \in \Omega ; \varphi(x) \neq 0\},$$

and the notation $\text{supp}(\varphi) \Subset \Omega$ means that $\text{supp}(\varphi)$ is a compact subset of Ω .

Equivalently, a function $\varphi \in C_c^\infty(\Omega)$ vanishes in a neighbourhood of the boundary $\partial\Omega$.

The vector

$$Du = (\partial_1 u, \dots, \partial_n u)$$

is called the *weak gradient* of u . It satisfies

$$\int_{\Omega} u \operatorname{div} \varphi \, dx = - \int_{\Omega} Du \cdot \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\Omega, \mathbb{R}^d). \quad (1.4)$$

Examples

The function $|x|$ in one dimension Consider $\Omega = (-1, 1) \subset \mathbb{R}$ and the function

$$u(x) = |x|.$$

This function is not classically differentiable at $x = 0$, but carry a weak derivative. Indeed, its weak derivative is given by

$$u'(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0, \end{cases}$$

which is integrable on $(-1, 1)$. We note here that u' is used here for the weak derivative in 1D.

To check this rigorously, let $\varphi \in C_c^\infty(\Omega)$ be a test function. By integration by parts, we have

$$\int_{-1}^1 |x| \varphi'(x) \, dx = \int_{-1}^0 (-x) \varphi'(x) \, dx + \int_0^1 x \varphi'(x) \, dx = - \int_{-1}^1 u'(x) \varphi(x) \, dx.$$

This shows that u' is indeed the weak derivative of u , and therefore u is weakly differentiable.

Compatibility with classical derivatives If $u \in C^1(\bar{\Omega})$, then the weak gradient Du exists and coincides with the classical gradient ∇u :

$$Du = \nabla u.$$

In other words, the notion of weak derivative extends the classical derivative: every classically differentiable function has a weak derivative that agrees with its classical gradient.

Remark: Most references denote the weak derivative simply by ∇u . In this chapter, we make a distinction between the classical gradient and the weak gradient for clarity.

Basic algebraic properties Let $u, v \in L^1(\Omega)$ admit weak derivatives. Let $\alpha, \beta \in \mathbb{R}$. Then the weak derivative is linear:

$$D(\alpha u + \beta v) = \alpha Du + \beta Dv,$$

where Du and Dv denote the weak derivatives of u and v .

If, in addition, $u, v \in L^1(\Omega) \cap L^\infty(\Omega)$ admit weak derivatives, then the product uv also admits a weak derivative, and

$$D(uv) = u Dv + v Du.$$

Uniqueness of weak derivatives Let $u \in L^1(\Omega)$ and assume that there exist $v, w \in L^1(\Omega)$ satisfying

$$\int_{\Omega} u \partial_i \varphi \, dx = - \int_{\Omega} v \varphi \, dx \quad \text{and} \quad \int_{\Omega} u \partial_i \varphi \, dx = - \int_{\Omega} w \varphi \, dx$$

for all test functions $\varphi \in C_c^\infty(\Omega)$. Then for all $\varphi \in C_c^\infty(\Omega)$,

$$\int_{\Omega} (v - w) \varphi \, dx = 0.$$

By the *fundamental lemma of the calculus of variations* (or standard result in measure theory), this implies

$$v = w$$

except on a set of zero measure.

Chain rule Let $u \in L^1(\Omega)$ admit a weak derivative, and let $f \in C^1(\mathbb{R})$ have a bounded derivative. Then the composition $f \circ u$ admits a weak derivative, and

$$D(f(u)) = f'(u) Du,$$

where Du denotes the weak derivative of u .

Remarks:

- This shows that weak derivatives are stable under smooth nonlinear transformations. For example, if $u(x) = |x|^\alpha$ admits a weak derivative, then functions like $\sin(u)$ or other smooth bounded functions of u also admit weak derivatives as long as the derivative of the outer function is bounded. This illustrates that functions with weak derivatives can have singularities or discontinuities while still being bounded.
- Positive and negative parts, $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$, admit weak derivatives whenever u does. This is often useful in variational problems and PDEs, since it allows working with truncated or sign-restricted functions. This property generally fails for classical C^1 functions.

Not all functions are weakly differentiable The Heaviside function

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0, \end{cases}$$

does not admit a weak derivative, because any candidate weak derivative would have to satisfy

$$\int_{\mathbb{R}} H(x) \varphi'(x) dx = - \int_{\mathbb{R}} H'(x) \varphi(x) dx$$

for all test functions $\varphi \in C_c^\infty(\mathbb{R})$. However, the left-hand side gives $\varphi(0)$, which cannot be represented as $\int f \varphi$ with $f \in L^2(\mathbb{R})$. Thus, H is not weakly differentiable. However one can give a distributional meaning to ∇H .

We record here that with the same line of arguments one can show that $u(x) = |x|, \in (-1, 1)$ is not twice weakly differentiable.

1.2.2 Distributions (additional material)

Test functions Let $\Omega \subset \mathbb{R}^n$ be open. We denote by $C_c^\infty(\Omega)$ the space of smooth functions with compact support in Ω , called *test functions*.

Distributions A *distribution* on Ω is a continuous linear functional on $C_c^\infty(\Omega)$. The space of distributions is denoted by $\mathcal{D}'(\Omega)$.

Every locally integrable function $u \in L_{\text{loc}}^1(\Omega)$ defines a distribution via

$$\langle u, \varphi \rangle = \int_{\Omega} u \varphi dx, \quad \varphi \in C_c^\infty(\Omega).$$

Distributional derivatives If $T \in \mathcal{D}'(\Omega)$, its derivative $\partial_i T$ is defined by

$$\langle \partial_i T, \varphi \rangle = -\langle T, \partial_i \varphi \rangle, \quad \varphi \in C_c^\infty(\Omega).$$

Weak derivatives are precisely distributional derivatives that are represented by functions in $L_{\text{loc}}^1(\Omega)$.

The Dirac distribution For a point $x_0 \in \Omega$, the *Dirac distribution* δ_{x_0} is defined by

$$\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0), \quad \varphi \in C_c^\infty(\Omega).$$

It is a distribution that cannot be represented by a locally integrable function, and plays a fundamental role in the theory of distributions and partial differential equations.

Derivative of the Heaviside The derivative of the Heaviside function $H(x)$ can be defined in the sense of distributions. Indeed, for all test functions $\varphi \in C_c^\infty(\mathbb{R})$, we have

$$\int_{\mathbb{R}} H(x) \varphi'(x) dx = -\varphi(0),$$

which suggests defining the distributional derivative of H as

$$\partial_x H = \delta_0,$$

where δ_0 is the Dirac distribution at the origin.

In other words, although H does not admit a classical or weak derivative as a function in $L^1(\mathbb{R})$, its derivative exists as a distribution. This illustrates how the framework of distributions extends the concept of derivative to functions with jumps or other singularities, which is particularly useful in PDEs and variational problems such as $-\Delta u = \delta_0$.

1.3 Linear Functional Spaces

Functional analysis views functions as points in normed vector spaces. In this framework, two functions u and v are considered close if the norm $\|u - v\|$ is small, meaning that they are similar with respect to the chosen notion of measurement. The advantage of this perspective is that one can choose a norm based on specific needs. We review several classical examples of function spaces.

Normed vector spaces A normed vector space is a vector space X endowed with a map $\|\cdot\| : X \rightarrow [0, \infty)$ satisfying, for all $u, v \in X$ and all $\lambda \in \mathbb{R}$,

$$\|u\| = 0 \iff u = 0, \quad \|\lambda u\| = |\lambda| \|u\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

The last property is the triangle inequality.

Convergence Let $(X, \|\cdot\|)$ be a normed vector space. A sequence $(u_n)_{n \in \mathbb{N}}$ in X is said to converge to $u \in X$ if

$$\|u_n - u\| \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

In this case, we write $u_n \rightarrow u$ in X .

This notion of convergence depends on the chosen norm. Different norms may lead to different concepts of closeness and therefore to different types of convergence.

A sequence (u_n) is called a Cauchy sequence if

$$\|u_n - u_m\| \longrightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

A normed space is said to be complete if every Cauchy sequence converges to a limit in X .

Spaces of continuous functions Let $\Omega \subset \mathbb{R}^n$ be open.

- $C^0(\overline{\Omega})$: the space of continuous functions on $\overline{\Omega}$, endowed with the uniform norm

$$\|u\|_{L^\infty(\Omega)} = \sup_{x \in \Omega} |u(x)|.$$

- $C^1(\overline{\Omega})$: the space of continuously differentiable functions on $\overline{\Omega}$, with norm

$$\|u\|_{C^1} = \|u\|_{L^\infty(\Omega)} + \|\nabla u\|_{L^\infty(\Omega)}.$$

- $C^{0,\alpha}(\overline{\Omega})$, $0 < \alpha \leq 1$: the space of functions in $C^0(\overline{\Omega})$ that are Hölder continuous, endowed with the norm

$$\|u\|_{C^{0,\alpha}} = \|u\|_{C^0} + \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha}.$$

- $C^{1,\alpha}(\overline{\Omega})$, $0 < \alpha \leq 1$: the space of functions in $C^1(\overline{\Omega})$ whose first derivatives are Hölder continuous, endowed with the norm

$$\|u\|_{C^{1,\alpha}} = \|u\|_{C^1} + \sup_{x \neq y} \frac{|\nabla u(x) - \nabla u(y)|}{|x - y|^\alpha}.$$

These spaces encode increasingly strong notions of regularity. From each of these spaces, one can also consider the subspaces of functions with compact support, denoted $C_c^k(\Omega)$ or $C_c^{k,\alpha}(\Omega)$, which consist of functions in the corresponding space that vanish outside a compact subset of Ω .

Lebesgue spaces $L^p(\Omega)$ For $1 \leq p < \infty$, the Lebesgue space $L^p(\Omega)$ consists of (measurable) functions such that

$$\int_{\Omega} |u|^p dx < \infty.$$

It is endowed with the norm

$$\|u\|_{L^p(\Omega)} = \left(\int_{\Omega} |u|^p dx \right)^{1/p}. \quad (1.5)$$

For $p = \infty$, one defines

$$\|u\|_{L^\infty(\Omega)} = \inf \left\{ M \geq 0 \mid \text{meas}(\{x \in \Omega : |f(x)| > M\}) = 0 \right\}.$$

We do not insist here on measurability issues nor on the fine properties of the Lebesgue integral, which are treated elsewhere.

Norm versus seminorm The quantity $(\int_{\Omega} |u|^p)^{1/p}$ does not distinguish between functions that differ only on sets of measure zero. For instance, the function

$$u(x) = \begin{cases} 1, & x = x_0, \\ 0, & x \neq x_0, \end{cases}$$

satisfies $\|u\|_{L^p(\Omega)} = 0$ for all $1 \leq p < \infty$.

Thus, two points of view are possible:

- identify functions that coincide almost everywhere, obtaining a true normed space,
- or keep pointwise-defined functions and interpret $\|\cdot\|_{L^p}$ as a seminorm.

In practice, there is no universal consensus on the “best” point of view. We present both, since the goal of this course is to enable you to read research papers and standard documentation, where either convention may be used.

What matters is the implication

$$\int_{\Omega} |u|^p dx = 0 \implies \text{meas}(\{x \in \Omega : u(x) \neq 0\}) = 0$$

Example of L^1 function: $u(x) = \chi_K(x)$ Let $\Omega \subset \mathbb{R}^N$ be an open set and let $K \subset \Omega$ be a compact set. Define

$$u(x) = \chi_K(x) = \begin{cases} 1 & \text{if } x \in K, \\ 0 & \text{if } x \notin K. \end{cases}$$

Since K is compact, it has finite Lebesgue measure. Hence

$$\int_{\Omega} |u(x)| dx = \int_{\Omega} \chi_K(x) dx = |K| < \infty.$$

Therefore $u \in L^1(\Omega)$.

More generally, for every $1 \leq p < \infty$,

$$\int_{\Omega} |u(x)|^p dx = \int_{\Omega} \chi_K(x) dx = |K| < \infty,$$

so $\chi_K \in L^p(\Omega)$ for all $1 \leq p < \infty$.

This example shows that functions with bounded support and finite measure are automatically integrable.

Example of L^1 function: $u(x) = 1/\sqrt{x}$ on $(0, 1)$ Let $\Omega = (0, 1)$ and define

$$u(x) = \frac{1}{\sqrt{x}}.$$

Then $u \in L^1(\Omega)$ but $u \notin L^2(\Omega)$.

General example: power functions $|x|^\alpha$ in $\mathbb{R}^N$ Let $\Omega = B(0, 1) \subset \mathbb{R}^N$ and consider

$$u(x) = |x|^\alpha.$$

Using polar coordinates, the integral of $|u|^p$ becomes

$$\int_{\Omega} |u(x)|^p dx = \int_0^1 r^{\alpha p} r^{N-1} dr \int_{\mathbb{S}^{N-1}} d\sigma = |\mathbb{S}^{N-1}| \int_0^1 r^{\alpha p + N-1} dr,$$

where $|\mathbb{S}^{N-1}|$ denotes the surface area of the unit sphere in $\mathbb{R}^N$.

The integral converges if and only if

$$\alpha p + N > 0 \quad \Longleftrightarrow \quad \alpha > -\frac{N}{p}.$$

Hence,

$$|x|^\alpha \in L^p(B(0, 1)) \quad \Longleftrightarrow \quad \alpha > -\frac{N}{p}.$$

This illustrates how integrability depends both on the singularity of the function at the origin and on the dimension N .

Interpolation and Hölder inequality A fundamental tool in the study of L^p spaces is Hölder's inequality: if $1 \leq p, q \leq \infty$ satisfy $1/p + 1/q = 1$, then

$$\int_{\Omega} |uv| dx \leq \|u\|_{L^p(\Omega)} \|v\|_{L^q(\Omega)}.$$

As a consequence, if $1 \leq p < r < q \leq \infty$, the space $L^r(\Omega)$ can be viewed as an intermediate space between $L^p(\Omega)$ and $L^q(\Omega)$. More precisely, one has interpolation inequalities of the form

$$\|u\|_{L^r(\Omega)} \leq \|u\|_{L^p(\Omega)}^\theta \|u\|_{L^q(\Omega)}^{1-\theta} \text{ provided } \frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}$$

for suitable $\theta \in (0, 1)$.

$L^2(\Omega) \subset L^1(\Omega)$ for bounded domains If Ω has finite measure $|\Omega| < \infty$, Hölder's inequality gives, for any $u \in L^2(\Omega)$,

$$\|u\|_{L^1(\Omega)} = \int_{\Omega} |u| dx \leq \left(\int_{\Omega} |u|^2 dx \right)^{1/2} \left(\int_{\Omega} 1^2 dx \right)^{1/2} = \|u\|_{L^2(\Omega)} \text{meas}(\Omega)^{1/2}.$$

This shows that

$$L^2(\Omega) \subset L^1(\Omega),$$

with a continuous embedding.

$L^2(\Omega)$ as a Hilbert space A vector space endowed with an inner product is called a *pre-Hilbert space*. If it is complete with respect to the induced norm, it is called a *Hilbert space*.

Inner product and Cauchy–Schwarz inequality Let H be a real vector space. An *inner product* on H is a map

$$\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{R}$$

such that for all $u, v, w \in H$ and all $\lambda \in \mathbb{R}$:

$$\langle u, v \rangle = \langle v, u \rangle, \quad \langle \lambda u + w, v \rangle = \lambda \langle u, v \rangle + \langle w, v \rangle,$$

and

$$\langle u, u \rangle \geq 0, \quad \langle u, u \rangle = 0 \iff u = 0.$$

The norm induced by the inner product is defined by

$$\|u\| := \sqrt{\langle u, u \rangle}.$$

A fundamental result is the *Cauchy–Schwarz inequality*:

$$|\langle u, v \rangle| \leq \|u\| \|v\| \quad \text{for all } u, v \in H.$$

In particular, this inequality shows that the inner product is always finite and continuous with respect to the induced norm.

The space $L^2(\Omega)$ is a Hilbert space when endowed with the inner product

$$\boxed{\langle u, v \rangle_{L^2} = \int_{\Omega} u(x)v(x) dx.} \tag{1.6}$$

The associated norm satisfies

$$\|u\|_{L^2} = \sqrt{\langle u, u \rangle_{L^2}}.$$

Typical examples of Hilbert spaces include:

- $\mathbb{R}^N$ with the Euclidean inner product

$$\langle x, y \rangle = \sum_{i=1}^N x_i y_i, \quad x, y \in \mathbb{R}^N.$$

- The space ℓ^2 of square-summable sequences, defined by

$$\ell^2 := \left\{ (x_n)_{n \geq 1} \subset \mathbb{R} \mid \sum_{n=1}^{\infty} |x_n|^2 < \infty \right\},$$

with the inner product

$$\langle (x_n), (y_n) \rangle_{\ell^2} := \sum_{n=1}^{\infty} x_n y_n.$$

- The Lebesgue space $L^2(\Omega)$, consisting of square-integrable functions on a domain Ω , with inner product

$$\langle u, v \rangle_{L^2(\Omega)} := \int_{\Omega} u(x) v(x) dx.$$

- Certain Sobolev spaces, such as $H^1(\Omega)$, which will be discussed later, equipped with the inner product

$$\langle u, v \rangle_{H^1(\Omega)} := \int_{\Omega} uv + \nabla u \cdot \nabla v dx.$$

Among Lebesgue spaces, $L^p(\Omega)$ is a Hilbert space if and only if $p = 2$.

Parallelogram identity In Hilbert spaces, the norm induced by the inner product satisfies the *parallelogram identity*:

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2, \quad (1.7)$$

for all u, v in the space.

This identity characterizes norms that arise from an inner product. In particular, a normed vector space is induced by an inner product if and only if the parallelogram identity holds.

Why functional analysis matters This illustrates the power of functional analysis: once a property is proved in the abstract setting of Hilbert spaces, it automatically applies to all concrete examples we encounter, such as $\mathbb{R}^N$, $L^2(\Omega)$, ℓ^2 , and Sobolev spaces.

1.4 Some Sobolev Spaces

Sobolev spaces provide a natural framework for variational problems involving derivatives. They combine integrability properties with weak differentiability and play a central role in the study of partial differential equations.

1.4.1 The Sobolev space $H^1(\Omega)$

Definition Let $\Omega \subset \mathbb{R}^n$ be an open set. The Sobolev space $H^1(\Omega)$ is defined as

$$H^1(\Omega) = \{u \in L^2(\Omega) : Du \in L^2(\Omega; \mathbb{R}^n)\}, \quad (1.8)$$

where Du denotes the weak gradient of u , introduced in the previous section.

Norm and inner product The space $H^1(\Omega)$ is endowed with the norm

$$\|u\|_{H^1(\Omega)} = \left(\int_{\Omega} |u|^2 dx + \int_{\Omega} |Du|^2 dx \right)^{\frac{1}{2}};$$

which is induced by the inner product

$$\langle u, v \rangle_{H^1} = \int_{\Omega} uv dx + \int_{\Omega} Du \cdot Dv dx$$

Other norms and inner product are available (see Lax-Milgram).

Completeness The space $H^1(\Omega)$, endowed with the norm $\|\cdot\|_{H^1}$, is a Hilbert space. In particular, it is complete: every Cauchy sequence in $H^1(\Omega)$ converges to a limit in $H^1(\Omega)$. This is a theorem that is proven elsewhere.

This property makes $H^1(\Omega)$ the natural energy space for the Dirichlet integral and for many elliptic boundary value problems.

Examples of Sobolev functions

Smooth functions Every function $u \in C^1(\overline{\Omega})$ belongs to $H^1(\Omega)$. Indeed, the weak derivative coincides with the classical gradient, and moreover $u \in L^2(\Omega)$ since Ω has finite measure. We have

$$\int_{\Omega} |Du|^2 dx = \int_{\Omega} |\nabla u|^2 dx \leq \text{meas}(\Omega) \|\nabla u\|_{L^\infty(\Omega)}^2,$$

and similarly

$$\int_{\Omega} |u|^2 dx \leq \text{meas}(\Omega) \|u\|_{L^\infty(\Omega)}^2.$$

Thus, every $u \in C^1(\overline{\Omega})$ belongs to $H^1(\Omega)$, and we have the inclusion

$$C^1(\overline{\Omega}) \subset H^1(\Omega).$$

The function $|x|$ in one dimension Let $\Omega = (-1, 1) \subset \mathbb{R}$ and consider

$$u(x) = |x|.$$

Although u is not classically differentiable at $x = 0$, it belongs to $H^1(\Omega)$. Its weak derivative is

$$u'(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0, \end{cases}$$

which is square-integrable on $(-1, 1)$. Indeed, we have

$$\int_{-1}^1 |u'(x)|^2 dx = \int_{-1}^0 (-1)^2 dx + \int_0^1 1^2 dx = 1 + 1 = 2.$$

Thus, $u \in H^1((-1, 1))$.

Power functions $|x|^\alpha$ in $\mathbb{R}^N$ and weak differentiability Let $\Omega = B(0, 1) \subset \mathbb{R}^N$ and consider

$$u(x) = |x|^\alpha.$$

Using polar coordinates, one finds

$$u \in L^p(\Omega) \iff \alpha p > -N,$$

and

$$u \in H^1(\Omega) \iff \alpha > 1 - \frac{N}{2}.$$

Hence, $|x|^\alpha \in H^1(B(0, 1))$ requires a stronger condition than L^2 -integrability. This illustrates that weak differentiability depends both on the singularity of the function and the dimension N of the domain. Functions not classically differentiable everywhere may still belong to H^1 if their weak derivatives are integrable.

Remarks on higher dimensions When $N \geq 2$, the range of α for which $|x|^\alpha \in H^1(B(0, 1))$ becomes more restrictive:

$$\begin{cases} N = 2 : & \alpha > 0, \\ N = 3 : & \alpha > -\frac{1}{2}. \end{cases}$$

This highlights that weak differentiability is sensitive to both the function's singularity and the ambient dimension. Note also that H^1 functions can be unbounded.

Restriction to a subdomain Let $\Omega_1 \subset \Omega_2 \subset \mathbb{R}^n$ be open sets. If

$$u \in H^1(\Omega_2),$$

then the restriction of u to Ω_1 belongs to $H^1(\Omega_1)$, that is,

$$u|_{\Omega_1} \in H^1(\Omega_1).$$

Moreover, the weak derivatives of $u|_{\Omega_1}$ are simply the restrictions of the weak derivatives of u on Ω_2 :

$$D(u|_{\Omega_1}) = (Du)|_{\Omega_1}.$$

Remark. This property shows that H^1 is *local* in the sense that membership and weak derivatives are preserved when restricting to smaller open sets.

1.5 Embeddings

All dimensions Any function $u \in H^1(\Omega)$ automatically belongs to $L^2(\Omega)$. Moreover, the L^2 -norm is controlled by the H^1 -norm:

$$\|u\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)} := \left(\|u\|_{L^2(\Omega)}^2 + \|Du\|_{L^2(\Omega)}^2 \right)^{1/2}.$$

In other words, we have

$$H^1(\Omega) \subset L^2(\Omega).$$

In dimension one In one dimension, functions in $H^1(\Omega)$ are continuous, although they may fail to be C^1 . For example, for

$$u(x) = |x|,$$

one can write, for any $x, y \in \Omega$,

$$u(x) - u(y) = \int_y^x u'(t) dt,$$

where u' is understood as the weak derivative.

By taking the absolute value and using the Cauchy–Schwarz inequality, we deduce

$$|u(x) - u(y)| \leq \int_y^x |u'(t)| dt \leq |x - y|^{1/2} \|u'\|_{L^2(\Omega)},$$

which gives a quantitative estimate of the variation of u in terms of its H^1 -norm. This illustrates the fundamental theorem for Sobolev functions in dimension 1.

In contrast, in higher dimensions, continuity is no longer guaranteed. Functions such as $|x|^\alpha$ or $\sin(|x|^\alpha)$ may be bounded but not continuous if α is too small.

Poincaré inequality

We assume smooth bounded domain Ω .

Gradient control For functions $u \in H_0^1(\Omega)$ that vanishes on the boundary, the classical Poincaré inequality states

$$\|u\|_{L^2(\Omega)} \leq C \|Du\|_{L^2(\Omega)},$$

where $C > 0$ depends only on the domain Ω . We define further in the text the space $H_0^1(\Omega)$; however we can say that it consists of all limits in $H^1(\Omega)$ of functions $C_c^1(\Omega)$ so that on the boundary in some sense the functions of $H_0^1(\Omega)$ vanishes.

Importantly $C = c \text{diam} \Omega$.

With mean For general $u \in H^1(\Omega)$, one has

$$\|u - \bar{u}\|_{L^2(\Omega)} \leq C \|Du\|_{L^2(\Omega)}, \quad \bar{u} = \frac{1}{|\Omega|} \int_{\Omega} u,$$

which is useful for Neumann-type problems.

Sobolev embeddings in dimension $N \geq 3$

The embedding

$$H^1(\Omega) \subset L^p(\Omega)$$

holds for $p = p^*$ with

$$p^* = \frac{2N}{N-2} \geq 3$$

Why this p^* Consider $u \in H^1(\mathbb{R}^N)$ and define the rescaled function

$$u_\lambda(x) = u(\lambda x), \quad \lambda > 0.$$

A scaling argument gives the critical Sobolev exponent p^* . Indeed, note that

$$\|u_\lambda\|_{L^p(\mathbb{R}^N)}^p = \int_{\mathbb{R}^N} |u(\lambda x)|^p dx = \lambda^{-N} \int_{\mathbb{R}^N} |u(y)|^p dy = \lambda^{-N} \|u\|_{L^p(\mathbb{R}^N)}^p,$$

while

$$\|Du_\lambda\|_{L^2(\mathbb{R}^N)}^2 = \int_{\mathbb{R}^N} |D(u(\lambda x))|^2 dx = \lambda^2 \lambda^{-N} \int_{\mathbb{R}^N} |Du(y)|^2 dy = \lambda^{2-N} \|Du\|_{L^2(\mathbb{R}^N)}^2.$$

To have a scale-invariant inequality $\|u\|_{L^p(\mathbb{R}^N)} \leq C \|Du\|_{L^2(\mathbb{R}^N)}$, one must have $\lambda^{-N/p} \sim \lambda^{(2-N)/2}$, which gives

$$p^* = \frac{2N}{N-2} \geq 2.$$

Sobolev embeddings in dimension $N = 2$

In $N = 2$, the function

$$u(x) = \log\left(\frac{\log|x|^2}{2}\right)$$

belongs to $H^1(B(0,1))$ but is not continuous at $x = 0$, neither bounded.

1.5.1 The Space $H^2(\Omega)$

Definition Let $\Omega \subset \mathbb{R}^n$ be an open set. The Sobolev space $H^2(\Omega)$ is defined as

$$H^2(\Omega) = \{u \in H^1(\Omega) : \text{Each component of } Du \text{ is in } H^1(\Omega)\},$$

Norm and inner product The natural norm on $H^2(\Omega)$ is

$$\|u\|_{H^2(\Omega)}^2 = \int_{\Omega} |u|^2 dx + \int_{\Omega} |\nabla u|^2 dx + \int_{\Omega} |D^2 u|^2 dx,$$

where $|\nabla u|^2 = \sum_{i=1}^n |\partial_i u|^2$ and $|D^2 u|^2 = \sum_{i,j=1}^n |\partial_i \partial_j u|^2$.

The corresponding inner product is

$$\langle u, v \rangle_{H^2} = \int_{\Omega} uv dx + \int_{\Omega} \nabla u \cdot \nabla v dx + \int_{\Omega} D^2 u : D^2 v dx,$$

where $D^2 u : D^2 v = \sum_{i,j=1}^n \partial_i \partial_j u \partial_i \partial_j v$.

Elliptic regularity and Fourier approach Let $u \in L^2(\Omega)$ satisfy $\Delta u \in L^2(\Omega)$ with suitable boundary conditions. Then $u \in H^2(\Omega)$.

In $\Omega = \mathbb{R}^n$, using the Fourier transform:

$$\widehat{\Delta u}(\xi) = -|\xi|^2 \hat{u}(\xi) \Rightarrow \int_{\mathbb{R}^n} |\xi|^4 |\hat{u}(\xi)|^2 d\xi = \|\Delta u\|_{L^2}^2 < \infty,$$

so all second derivatives of u are square-integrable, with

$$\|D^2 u\|_{L^2} \leq \|\Delta u\|_{L^2}, \quad \|u\|_{H^2} \leq C(\|u\|_{L^2} + \|\Delta u\|_{L^2}).$$

This shows that for $u \in H^1(\Omega)$

$$\Delta u \in L^2(\Omega) \Rightarrow u \in H^2(\Omega).$$

Regularity theory To obtain smooth solutions, one first constructs them in the Sobolev space $H^1(\Omega)$. The previous observation shows that if

$$\Delta u = f \in L^2(\Omega),$$

then $u \in H^2(\Omega)$, which is a stronger regularity than the a priori H^1 regularity.

Symmetry of second derivatives: Schwarz's lemma For functions in $H^2(\Omega)$, the weak second derivatives satisfy the symmetry property

$$\partial_i \partial_j u = \partial_j \partial_i u \quad \text{in } L^2(\Omega),$$

for all $i, j = 1, \dots, N$. This is the weak version of Schwarz's lemma (also called Clairaut's theorem). This shows that the Hessian matrix D^2u is symmetric almost everywhere, which is crucial in applications such as plate theory and fourth-order PDEs.

Sobolev embeddings and continuity for H^2 functions Just as H^1 functions enjoy improved integrability compared to L^2 functions, H^2 regularity provides even stronger properties. In particular, $H^2(\Omega)$ embeds into spaces of continuous or Hölder-continuous functions in low dimensions:

- **Dimension $N = 1$:** The embedding $H^2(\Omega) \subset C^1(\overline{\Omega})$ holds; functions are continuously differentiable.
- **Dimension $N = 2$:** We have $H^2(\Omega) \subset C^{0,\alpha}(\overline{\Omega})$ for every $\alpha \in (0, 1)$; functions are Hölder continuous (hence continuous).
- **Dimension $N = 3$:** The embedding $H^2(\Omega) \subset C^{0,\alpha}(\overline{\Omega})$ holds with $\alpha = \frac{1}{2}$; functions are Hölder continuous but not necessarily differentiable.
- **Dimension $N = 4$:** One has $H^2(\Omega) \subset L^p(\Omega)$ for all $p < \infty$, but no continuity can be inferred.
- **Dimension $N \geq 5$:** The embedding $H^2(\Omega) \subset L^p(\Omega)$ holds for $p \leq p^{**} := \frac{2N}{N-4}$; continuity is generally lost.

These embeddings follow from iterated application of Sobolev embedding theorems. For instance, in dimension 2, we have $H^2 \subset H^1 \subset L^p$ for any $p < \infty$, and by the Rellich-Kondrachov theorem, this implies $H^2 \subset C^0$ in bounded domains. In dimension 1, one can show by direct integration that H^2 functions have continuous derivatives. These results are particularly important for fourth-order problems such as the biharmonic equation $\Delta^2 u = f$, where H^2 is the natural energy space.

Remark: In the context of plate deformation (biharmonic problem), this explains why the displacement u is continuous in dimensions 1, 2, and 3: the H^2 regularity of u ensures at least continuity (or Hölder continuity) of the solution in these dimensions.

1.5.2 Interpretation in functional analysis

All these inequalities can be understood in terms of functional analysis.

Continuous linear applications A map $L : (X, \|\cdot\|_X) \rightarrow (Y, \|\cdot\|_Y)$ between normed spaces is *linear and continuous* if it is linear and there exists a constant $C > 0$ such that

$$\|L(u)\|_Y \leq C\|u\|_X \quad \text{for all } u \in X.$$

By linearity

$$\|L(u) - L(v)\|_Y \leq C\|u - v\|_X.$$

Example: integration as a continuous linear map If Ω has finite measure, the map

$$I : L^1(\Omega) \rightarrow \mathbb{R}, \quad u \mapsto \int_{\Omega} u(x) dx$$

is linear and continuous, since

$$\left| \int_{\Omega} u(x) dx \right| \leq \int_{\Omega} |u(x)| dx = \|u\|_{L^1(\Omega)}.$$

Here the constant is $C = 1$.

Example: use Sobolev embedding in dimension 3 Let $\Omega \subset \mathbb{R}^3$ be a bounded domain, $f \in L^{6/5}(\Omega)$, and $u \in H^1(\Omega)$. Define the linear map

$$L : H^1(\Omega) \rightarrow \mathbb{R}, \quad L(u) = \int_{\Omega} f(x) u(x) dx.$$

By the Sobolev embedding $H^1(\Omega) \subset L^6(\Omega)$ in dimension 3, we have

$$\|u\|_{L^6(\Omega)} \leq C \|u\|_{H^1(\Omega)}.$$

Using Hölder's inequality, we get

$$|L(u)| = \left| \int_{\Omega} f u dx \right| \leq \|f\|_{L^{6/5}(\Omega)} \|u\|_{L^6(\Omega)} \leq C \|f\|_{L^{6/5}(\Omega)} \|u\|_{H^1(\Omega)}.$$

Hence L is a continuous linear map on $H^1(\Omega)$, with operator norm controlled by $\|f\|_{L^{6/5}(\Omega)}$.

Example: reflection as a continuous linear operator Let $\Omega = (-1, 1)$ and define the reflection operator

$$R : L^2(-1, 1) \rightarrow L^2(-1, 1), \quad (Ru)(x) = u(-x).$$

The map R is linear. Moreover, it is continuous and preserves the L^2 norm. Indeed, by the change of variables $y = -x$, we compute

$$\|Ru\|_{L^2(-1,1)}^2 = \int_{-1}^1 |u(-x)|^2 dx = \int_{-1}^1 |u(y)|^2 dy = \|u\|_{L^2(-1,1)}^2.$$

Therefore,

$$\|Ru\|_{L^2(-1,1)} = \|u\|_{L^2(-1,1)},$$

and R is an isometry of $L^2(-1, 1)$.

In particular, R is a bounded linear operator with operator norm

$$\|R\|_{\mathcal{L}(L^2, L^2)} = 1.$$

Sobolev embeddings as linear operators Sobolev embeddings can be interpreted as linear continuous maps. For example, the embedding

$$H^1(\Omega) \hookrightarrow L^p(\Omega)$$

defines the identity map

$$\text{id} : H^1(\Omega) \rightarrow L^p(\Omega), \quad u \mapsto u,$$

which is linear and continuous:

$$\|u\|_{L^p(\Omega)} \leq C \|u\|_{H^1(\Omega)}.$$

Finite versus infinite dimension In finite-dimensional spaces, any linear map is automatically continuous. For example, let $x \in \mathbb{R}^N$ and let $A \in \mathbb{R}^{N \times N}$ be a square matrix. Then the linear map

$$L : \mathbb{R}^N \rightarrow \mathbb{R}^N, \quad L(x) = Ax,$$

is linear and continuous. Indeed, one can take the operator norm of A to bound $\|L(x)\| \leq \|A\| \|x\|$ for any vector norm on $\mathbb{R}^N$.

In contrast, in infinite-dimensional spaces, linear maps need not be continuous. Sobolev embedding inequalities provide explicit bounds ensuring continuity of natural linear operators, such as the identity or inclusion maps, in this context.

This illustrates how functional analysis allows us to interpret classical inequalities as statements about **continuous linear operators** between infinite-dimensional spaces, in analogy with matrices acting on finite-dimensional vectors.

As a side note $\Pi(u)$ is not a linear and continuous operator as it is not linear.

Korn's inequality as a continuous linear operator In linear elasticity, the symmetric gradient

$$\varepsilon : H^1(\Omega)^d \rightarrow L^2(\Omega)_{\text{sym}}^{d \times d}, \quad u \mapsto \varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T),$$

defines a linear operator.

Korn's inequality states that there exists a constant $C > 0$ such that

$$\|u\|_{H^1(\Omega)} \leq C(\|u\|_{L^2(\Omega)} + \|\varepsilon(u)\|_{L^2(\Omega)}) \quad \text{for all } u \in H^1(\Omega).$$

This inequality shows that the inverse of ε , restricted to $\mathcal{A}$, is a continuous linear operator. Equivalently, the strain norm

$$\|u\|_\varepsilon = \|\varepsilon(u)\|_{L^2(\Omega)}$$

is equivalent to the H^1 norm on $\mathcal{A}$.

1.6 Approximation Results

We present three fundamental approximation results, which can later be understood in terms of dense classes in functional spaces.

1.6.1 Three results

1. Truncation and bounded approximation Let $u \in H^1(\Omega)$. Define the truncated function

$$T_k(u) = \max(-k, \min(u, k)) = \begin{cases} k & u > k \\ u & u \in [-k, k] \\ -k & u < -k \end{cases}, \quad k > 0.$$

Then $T_k(u) \in H^1(\Omega) \cap L^\infty(\Omega)$ and

$$\|DT_k(u)\|_{L^2(\Omega)} \leq \|Du\|_{L^2(\Omega)}.$$

Moreover, $T_k(u) \rightarrow u$ in $H^1(\Omega)$ as $k \rightarrow \infty$, providing a way to approximate H^1 functions by bounded functions.

2. Piecewise linear approximation If Ω is a polygonal domain, functions in $H^1(\Omega)$ can be approximated by piecewise linear functions (e.g., finite element functions) in the H^1 norm. However, this approximation generally does **not** work in H^2 , because piecewise linear functions have discontinuous derivatives.

3. Smooth approximation For $u \in H^1(\mathbb{R}^d)$, let ρ_ϵ be a smooth mollifier. The convolution

$$u_\epsilon = \rho_\epsilon * u = \int_{\mathbb{R}^N} \rho_\epsilon(x-y)u(y)dy$$

yields a smooth function $u_\epsilon \in C^\infty(\Omega)$, and

$$u_\epsilon \rightarrow u \quad \text{in } H^1(\Omega) \text{ as } \epsilon \rightarrow 0.$$

This shows that smooth functions are dense in $H^1(\Omega)$. Historically, this motivates the identification $H^1(\Omega) = W^{1,2}(\Omega)$ as introduced in functional analysis literature [5].

1.6.2 Dense classes

Definition of density A subset $D \subset X$ of a normed space $(X, \|\cdot\|)$ is *dense* if for every $u \in X$ there exists a sequence $(u_n)_{n \geq 1} \subset D$ such that

$$u_n \rightarrow u \quad \text{in the norm of } X.$$

Rewriting the three approximation results in terms of dense classes

- **Truncation:** $H^1(\Omega) \cap L^\infty(\Omega)$ is dense in $H^1(\Omega)$.
- **Piecewise linear functions:** In a polygonal domain, the set of piecewise linear functions is dense in $H^1(\Omega)$.
- **Smooth functions:** $C^\infty(\Omega) \cap H^1(\Omega)$ is dense in $H^1(\Omega)$.

These dense classes are essential both for theoretical purposes (defining weak solutions) and practical purposes (numerical approximations, e.g., finite elements).

1.6.3 Definition of $H_0^1(\Omega)$ by density

Test functions Let $\Omega \subset \mathbb{R}^N$ be open. We denote by $C_c^\infty(\Omega)$ the space of smooth functions with compact support in Ω . These functions are often called *test functions*.

Definition of $H_0^1(\Omega)$ The Sobolev space $H_0^1(\Omega)$ is defined as the closure of $C_c^\infty(\Omega)$ in the H^1 norm:

$$H_0^1(\Omega) := \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{H^1}}.$$

Equivalently, $u \in H_0^1(\Omega)$ if there exists a sequence $(u_n)_{n \geq 1} \subset C_c^\infty(\Omega)$ such that

$$\|u_n - u\|_{H^1(\Omega)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This enables us to define functions vanishing on the boundary. The set $H_0^1(\Omega)$ with the inner product inherited from $H^1(\Omega)$ is a Hilbert space

1.7 Restriction of a Sobolev function: the trace

Motivation How can one give meaning to the values of a Sobolev function on the boundary of a domain, when the function is only defined up to sets of measure zero? The idea is to use *approximation by smooth functions* and interpret the trace as a *continuous linear map*. We define $\text{Tr}_{\partial\Omega} u = u|_{\partial\Omega}$ if $u \in C^1(\overline{\Omega})$

No problem in $N = 1$ If $\Omega \subset \mathbb{R}$, then $H^1(\Omega) \subset C^0(\bar{\Omega})$, so functions are continuous. The restriction to the endpoints (the "trace") is well-defined:

$$|\text{Tr}_{\partial\Omega} u(x)| \leq \|u\|_{C^0(\bar{\Omega})} \leq C\|u\|_{H^1(\Omega)}, \quad x \in \partial\Omega.$$

Trace in higher dimensions In dimension $N \geq 2$, the restriction of a general $L^2(\Omega)$ function to the boundary is ill-posed: there is no well-defined pointwise value on $\partial\Omega$.

For $u \in H^1(\Omega)$, the *trace operator*

$$\text{Tr}_{\partial\Omega} : H^1(\Omega) \rightarrow L^2(\partial\Omega)$$

is linear and continuous. More precisely, there exists a constant $C > 0$ such that

$$\|\text{Tr}_{\partial\Omega}(u)\|_{L^2(\partial\Omega)} \leq C\|u\|_{H^1(\Omega)}.$$

It is defined by density: if $(u_n) \subset C^\infty(\bar{\Omega})$ converges to u in $H^1(\Omega)$, then

$$\|\text{Tr}_{\partial\Omega}(u_n) - \text{Tr}_{\partial\Omega}(u_m)\|_{L^2(\partial\Omega)} \leq C\|u_n - u_m\|_{H^1(\Omega)} \rightarrow 0,$$

so $\text{Tr}_{\partial\Omega}(u)$ is well-defined, since $L^2(\partial\Omega)$ is complete.

Integration by parts formula Let $u \in H^1(\Omega)$ and $\varphi \in H^1(\Omega; \mathbb{R}^N)$. Then

$$\int_{\Omega} u \operatorname{div} \varphi \, dx = - \int_{\Omega} Du \cdot \varphi \, dx + \int_{\partial\Omega} (\text{Tr}_{\partial\Omega} u) (\varphi \cdot n) \, dS, \quad (1.9)$$

where n denotes the outward unit normal on $\partial\Omega$. For smooth $\varphi \in C^\infty(\bar{\Omega}; \mathbb{R}^N)$, this coincides with the classical integration by parts formula.

Poincaré inequality with non-zero boundary data Let $u, v \in H^1(\Omega)$ have the same trace g on $\partial\Omega$ ($u, v \in H_g^1(\Omega)$). Then

$$\|u - v\|_{L^2(\Omega)} \leq C\|Du - Dv\|_{L^2(\Omega)}.$$

This follows from the Poincaré inequality in $H_0^1(\Omega)$ applied to $u - v \in H_0^1(\Omega)$.

Gluing functions across subdomains If $\Omega_1, \Omega_2 \subset \mathbb{R}^N$ are two domains sharing a portion of their boundary, one can define a function on $\Omega_1 \cup \Omega_2$ by "gluing" functions $u_1 \in H^1(\Omega_1)$ and $u_2 \in H^1(\Omega_2)$ that agree on the common boundary portion. The integration by parts formula (1.9) ensures that the resulting function belongs to $H^1(\Omega_1 \cup \Omega_2)$, with a well-defined weak derivative. This property is not true for C^1 -functions.

Normal derivative trace for H^2 functions For functions in $H^2(\Omega)$, one can similarly define the trace of the *normal derivative* on the boundary. If $u \in H^2(\Omega)$ and Ω has a sufficiently regular boundary (e.g., C^1 or Lipschitz), the normal derivative

$$\partial_\nu u := \nabla u \cdot n$$

admits a trace on $\partial\Omega$, where n is the outward unit normal. More precisely, there exists a continuous linear operator

$$\text{Tr}_{\partial\Omega}(\partial_\nu \cdot) : H^2(\Omega) \rightarrow L^2(\partial\Omega)$$

such that

$$\|\text{Tr}_{\partial\Omega}(\partial_\nu u)\|_{L^2(\partial\Omega)} \leq C\|u\|_{H^2(\Omega)}.$$

This is again defined by density: for smooth functions $u \in C^2(\overline{\Omega})$, the normal derivative $\partial_\nu u = \nabla u \cdot n$ is well-defined on $\partial\Omega$, and the trace extends by continuity to all of $H^2(\Omega)$. This normal derivative trace is essential for formulating boundary conditions in fourth-order problems (such as plate bending equations) and for the Green formula associated with the biharmonic operator:

$$\int_{\Omega} \Delta u \Delta v \, dx = \int_{\Omega} D^2 u : D^2 v \, dx - \int_{\partial\Omega} \partial_\nu(\Delta u) v \, dS + \int_{\partial\Omega} (\Delta u) \partial_\nu v \, dS.$$

In particular, for $H^2(\Omega)$ functions, both $\text{Tr}_{\partial\Omega}(u)$ and $\text{Tr}_{\partial\Omega}(\partial_\nu u)$ are well-defined in $L^2(\partial\Omega)$, allowing one to impose Dirichlet and Neumann boundary conditions simultaneously, as required in clamped plate problems.

1.8 Lax–Milgram: the classical use of Hilbert spaces in FEM

A fundamental tool in the mathematical analysis of the finite element method is the Lax–Milgram theorem and its finite-dimensional counterpart. We begin with a general abstract result.

An abstract problem Let H be a Hilbert space and let

$$a : H \times H \rightarrow \mathbb{R}$$

be a bilinear form satisfying the continuity and coercivity estimates

$$|a(u, v)| \leq \alpha \|u\|_H \|v\|_H, \quad a(u, u) \geq \beta \|u\|_H^2, \quad (1.10)$$

for all $u, v \in H$, where $\alpha, \beta > 0$ are constants. Note that $\beta \leq \alpha$ whenever the bilinear form is continuous and coercive.

Let $L : H \rightarrow \mathbb{R}$ be a continuous linear functional.

We consider the problem

$$a(u, v) = L(v) \quad \text{for all } v \in H. \quad (1.11)$$

Lax–Milgram (coercive, not necessarily symmetric) The symmetry of a is *not* required for well-posedness.

If a is continuous and coercive in the sense of (1.10), then for every continuous linear functional L there exists a unique $u \in H$ such that (1.11) holds. Moreover, the solution satisfies the stability estimate

$$\|u\|_H \leq \frac{1}{\beta} \sup_{x \in H; \|x\| \leq 1} \|L(x)\|_H.$$

(1.12)

Variational interpretation in the symmetric case If a is symmetric, that is,

$$a(u, v) = a(v, u) \quad \text{for all } u, v \in H,$$

then the problem (1.11) admits a variational interpretation.

Indeed, in this case one can introduce the energy functional

$$\mathcal{J}(u) = \frac{1}{2} a(u, u) - L(u), \quad (1.13)$$

defined on H .

A function $u \in H$ is a solution of (1.11) if and only if it is a critical point of $\mathcal{J}$, that is,

$$\left. \frac{d}{dt} \mathcal{J}(u + tv) \right|_{t=0} = 0 \quad \text{for all } v \in H.$$

Moreover, by the coercivity assumption (1.10), the functional $\mathcal{J}$ is strictly convex and coercive. As a consequence, it admits a unique minimizer, which coincides with the solution of (1.11).

Thus, when a is symmetric, the Lax–Milgram problem can be interpreted as the minimization of an energy functional.

Also, when a is symmetric, we have that H with a is an Hilbert space.

Coercivity as a quantitative injectivity condition Coercivity can be interpreted as a quantitative form of injectivity.

In a finite-dimensional space, let $A \in \mathbb{R}^{N \times N}$ be a square matrix and define the bilinear form

$$a(u, v) := (Au, v) \quad \text{for } u, v \in \mathbb{R}^N,$$

where $(\cdot, \cdot)$ denotes the standard Euclidean inner product.

Then A is injective if and only if a is coercive, i.e., there exists $\beta > 0$ such that

$$a(u, u) = (Au, u) \geq \beta \|u\|^2 \quad \text{for all } u \in \mathbb{R}^N.$$

Remark. Coercivity not only ensures injectivity, but also gives a ****quantitative lower bound**** on how Au controls u , which is crucial in finite element analysis and stability estimates.

Existence, uniqueness and stability Under the assumptions (1.10), there exists a unique solution $u \in H$ of (1.11). This is the content of the Lax–Milgram theorem (1.12).

Galerkin approximation Let $V \subset H$ be a finite-dimensional subspace. The Galerkin approximation consists in finding $u_V \in V$ such that

$$\boxed{a(u_V, v) = L(v) \quad \text{for all } v \in V.} \tag{1.14}$$

This problem admits a unique solution $u_V \in V$.

If $\{\varphi_i\}_{i=1}^N$ is a basis of V , the Galerkin equations are equivalent to the linear system

$$A\mathbf{u} = \mathbf{f}, \quad A_{ij} := B(\varphi_j, \varphi_i), \quad \mathbf{f}_i := F(\varphi_i), \quad u_V = \sum_{j=1}^N \mathbf{u}_j \varphi_j.$$

If B is symmetric, then A is symmetric; if B is coercive, then the symmetric part $\frac{1}{2}(A + A^\top)$ is positive definite and A is invertible.

Using Galerkin orthogonality, the exact solution u and its Galerkin approximation u_V satisfy the error estimate

$$\boxed{\|u - u_V\|_H \leq \frac{\alpha}{\beta} \inf_{v \in V} \|u - v\|_H.} \tag{1.15}$$

This inequality is known as Céa’s lemma. It shows that the Galerkin solution is quasi-optimal: its error is controlled by the best possible approximation error in V .

1.8.1 Let’s use Lax–Milgram

We illustrate the use of the Lax–Milgram theorem through some classical examples of elliptic problems. The goal is to show existence and uniqueness of solutions of equations of the type

$$Au = f.$$

Example 1: Poisson problem with homogeneous Dirichlet boundary conditions Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with Lipschitz boundary. Consider the Poisson problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (1.16)$$

where $f \in L^2(\Omega)$.

We introduce the Hilbert space $H = H_0^1(\Omega)$ with the standard inner product

$$(u, v)_H := \int_{\Omega} \nabla u \cdot \nabla v \, dx, \quad \|u\|_H = \|\nabla u\|_{L^2(\Omega)}.$$

- **Continuity:** By Cauchy–Schwarz inequality,

$$|B(u, v)| \leq \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} = \|u\|_H \|v\|_H.$$

- **Coercivity:** With the chosen norm,

$$B(u, u) = \|\nabla u\|_{L^2}^2 = \|u\|_H^2,$$

so (1.10) holds with $\beta = 1$.

Example 1 (bis): Elliptic problem with variable coefficients and homogeneous Dirichlet boundary conditions Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with Lipschitz boundary. Consider the problem

$$\begin{cases} -\operatorname{div}(k(x)\nabla u) + q(x)u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (1.17)$$

where $f \in L^2(\Omega)$. We assume

$$k \in L^\infty(\Omega), \quad q \in L^\infty(\Omega),$$

and that there exist constants $k_0, k_1, q_0 > 0$ such that

$$0 < k_0 \leq k(x) \leq k_1 \quad \text{a.e. in } \Omega, \quad q(x) \geq q_0 \quad \text{a.e. in } \Omega.$$

We introduce the Hilbert space $H = H_0^1(\Omega)$ equipped with the equivalent norm

$$\|u\|_H^2 := \|\nabla u\|_{L^2(\Omega)}^2 + \|u\|_{L^2(\Omega)}^2.$$

The associated bilinear form is

$$B(u, v) := \int_{\Omega} k(x) \nabla u \cdot \nabla v \, dx + \int_{\Omega} q(x) uv \, dx.$$

- **Continuity:** By Cauchy–Schwarz inequality and boundedness of the coefficients,

$$\left| \int_{\Omega} k(x) \nabla u \cdot \nabla v \, dx \right| \leq k_1 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2},$$

$$\left| \int_{\Omega} q(x) uv \, dx \right| \leq \|q\|_{L^\infty} \|u\|_{L^2} \|v\|_{L^2}.$$

Therefore,

$$|B(u, v)| \leq C \|u\|_H \|v\|_H,$$

for some constant $C > 0$.

- **Coercivity:** For $u \in H_0^1(\Omega)$,

$$B(u, u) = \int_{\Omega} k(x) |\nabla u|^2 dx + \int_{\Omega} q(x) |u|^2 dx \geq k_0 \|\nabla u\|_{L^2}^2 + q_0 \|u\|_{L^2}^2.$$

Hence,

$$B(u, u) \geq \min(k_0, q_0) (\|\nabla u\|_{L^2}^2 + \|u\|_{L^2}^2) = \beta \|u\|_H^2,$$

with $\beta = \min(k_0, q_0) > 0$.

Remark: the quadratic form $\int_0^1 r |u'|^2 + \frac{u^2}{r} dr$ does not satisfy the coercivity assumptions due to the singular weight $1/r$ at $r = 0$.

Example 2: Poisson problem with Neumann boundary conditions Consider

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ \partial_n u = 0 & \text{on } \partial\Omega, \end{cases}$$

with $f \in L^2(\Omega)$ satisfying the compatibility condition $\int_{\Omega} f dx = 0$.

Take the quotient space $H = H^1(\Omega)/\mathbb{R}$ (functions modulo constants) with norm induced by the H^1 seminorm,

$$\|u\|_H := \|\nabla u\|_{L^2(\Omega)}, \quad B(u, v) = \int_{\Omega} \nabla u \cdot \nabla v dx, \quad F(v) = \int_{\Omega} f v dx.$$

Then B is continuous and coercive on H (by Poincaré inequality on mean-zero representatives), and Lax–Milgram ensures the existence of a unique solution in the quotient space, i.e., a solution **defined up to an additive constant**.

Example 3: Linear elasticity (vector-valued problem) Let $\Omega \subset \mathbb{R}^d$ and $u : \Omega \rightarrow \mathbb{R}^d$. The weak form of linear elasticity reads: find $u \in H_0^1(\Omega; \mathbb{R}^d)$ such that

$$\int_{\Omega} \sigma(u) : \varepsilon(v) dx = \int_{\Omega} f \cdot v dx, \quad \forall v \in H_0^1(\Omega; \mathbb{R}^d),$$

where $\varepsilon(u) = \frac{1}{2}(\nabla u + \nabla u^T)$ is the strain tensor and $\sigma(u)$ the stress tensor satisfying Hooke's law.

By Korn's inequality, $B(u, v) = \int_{\Omega} \sigma(u) : \varepsilon(v) dx$ is continuous and coercive on $H_0^1(\Omega; \mathbb{R}^d)$. Hence, Lax–Milgram guarantees a unique displacement u .

Remark on noncoercive problems When coercivity fails (e.g., pure convection, mixed formulations), the appropriate tool is the Banach–Nečas–Babuška (BNB) or inf-sup theory rather than Lax–Milgram.

Example 4 : Discrete Laplacians

Finite element (FE) stiffness matrix as a discrete Laplacian Let $\{\varphi_i\}_{i=1}^N$ be a nodal basis of a conforming FE space $V_h \subset H_0^1(\Omega)$. The FE *discrete Laplacian* (Dirichlet case) is the stiffness matrix

$$A_{ij} := \int_{\Omega} \nabla \varphi_j \cdot \nabla \varphi_i dx.$$

Then the Galerkin system reads $A\mathbf{u} = \mathbf{f}$ with $\mathbf{f}_i = \int_{\Omega} f \varphi_i dx$. The matrix A is symmetric and positive definite. For Neumann boundary conditions, the same assembly yields a symmetric *positive semidefinite* matrix with $\ker A = \text{span}\{\mathbf{1}\}$, reflecting the additive constant indeterminacy.

Example 5: Solving $Ax = b$ via Lax–Milgram Let $H = \mathbb{R}^n$ with the Euclidean inner product $(u, v) = v^\top u$. Given a matrix $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$, define

$$B(u, v) := (Au, v) = v^\top Au, \quad F(v) := (b, v) = v^\top b.$$

Then:

- **Continuity:** $|B(u, v)| \leq \|A\|_2 \|u\|_2 \|v\|_2$ with $\alpha = \|A\|_2$.
- **Coercivity:** If A is injective, there exists $\beta > 0$

$$v^\top Av \geq \beta \|v\|_2^2 \quad \forall v \in \mathbb{R}^n,$$

then Lax–Milgram ensures a unique u solving $B(u, v) = F(v)$ for all v , i.e., $Au = b$, and the stability estimate $\|u\|_2 \leq \frac{1}{\beta} \|b\|_2$.

Conclusion These examples show that the Lax–Milgram theorem provides a **systematic tool** to prove existence and uniqueness of solutions for linear PDEs in a variational framework and to justify the well-posedness of their Galerkin discretizations, including the structure of the resulting linear systems. However, in practice, many interesting problems fall outside this idealized setting: truly nonlinear or noncoercive problems cannot be handled directly by Lax–Milgram. Examples include the nonlinear Schrödinger equation, harmonic map equations, or many geometric variational problems. In such cases, the abstract Hilbert space framework only provides intuition or inspiration, but additional techniques (e.g., compactness methods, fixed point theorems, approximation) are needed to establish existence or uniqueness of solutions.

1.9 Exercises

Exercise 1.9.1 (Powers of $|x|$). *Let*

$$u(x) = |x|^\alpha \quad \text{on } B(0, 1),$$

where $\alpha \in \mathbb{R}$.

- Determine for which values of α the function u belongs to $L^2(B(0, 1))$.
- Determine for which values of α the function u belongs to $H^1(B(0, 1))$.
- Determine for which values of α the function u belongs to $H^2(B(0, 1))$.

Verify the inclusions between these spaces on the intervals you find.

Deduce that $|x|$ is not in $H^2(B(0, 1))$

Does your answer depend on the dimension if $u(x) = |x|^\alpha$ is defined on a ball in $\mathbb{R}^n$?

Hint. Study the integrability of u , Du , and D^2u near the singular point $x = 0$. Reduce the problem to the convergence of integrals of the form

$$\int_0^1 x^\beta dx.$$

Exercise 1.9.2 (Strict inclusions between Sobolev spaces). *Show that the inclusions*

$$H^2((-1, 1)) \subset H^1((-1, 1)) \subset L^2((-1, 1))$$

are strict.

Hint. Use the previous exercise to construct explicit functions that belong to one space but not to the next.

Exercise 1.9.3 (Piecewise linear functions and H^2). *Let $u : (-1, 1) \rightarrow \mathbb{R}$ be a piecewise linear function with at least one corner point.*

Show that $u \notin H^2((-1, 1))$.

Hint. Compute the weak second derivative of $u(x) = |x|$. Show that it contains Dirac masses at the corner points.

Exercise 1.9.4 (Embedding of L^2 into L^1). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain.*

Show that

$$L^2(\Omega) \subset L^1(\Omega),$$

and that this inclusion is continuous and strict.

Hint. Apply Hölder's inequality with $p = 2$ and $q = 2$, and use the finiteness of the measure of Ω .

Exercise 1.9.5 (Minimizing sequences in linear elasticity). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain and assume that $\Gamma_D = \partial\Omega$ has positive measure and $f \in L^2(\Omega)$.*

Let $\mathbb{C}$ be a uniformly positive definite elasticity tensor and consider the elastic energy

$$\Pi(u) = \int_{\Omega} \frac{1}{2} \varepsilon(u) : \mathbb{C} : \varepsilon(u) \, dx - \int_{\Omega} f \cdot u \, dx$$

defined on

$$\mathcal{A} = \{u \in H^1(\Omega)^d : u = 0 \text{ on } \Gamma_D\}.$$

Let $(u_n)_{n \in \mathbb{N}} \subset \mathcal{A}$ be a minimizing sequence:

$$\Pi(u_n) \longrightarrow \inf_{u \in \mathcal{A}} \Pi(u).$$

The goal of the exercise is to reproduce the introduction of the chapter.

1. *Show that (u_n) is a Cauchy sequence with respect to the strain norm*

$$\|u\|_{\varepsilon} = \|\varepsilon(u)\|_{L^2(\Omega)}.$$

2. *Using Korn's inequality, deduce that (u_n) is a Cauchy sequence in $H^1(\Omega)^d$.*

3. *Conclude that (u_n) converges in $H^1(\Omega)^d$ to a minimizer of Π .*

Exercise 1.9.6 (L^p and L^q inclusions for bounded domains). *Let $\Omega \subset \mathbb{R}^n$ be a measurable set with finite measure $|\Omega| < \infty$, and let $1 \leq q \leq p \leq \infty$.*

1. *Show that for any $u \in L^p(\Omega)$,*

$$\|u\|_{L^q(\Omega)} \leq \text{meas}(\Omega)^{\frac{1}{q} - \frac{1}{p}} \|u\|_{L^p(\Omega)}.$$

Conclude that

$$L^p(\Omega) \subset L^q(\Omega),$$

with a continuous linear operator.

2. *Show that the inclusion is strict.*

Hint. Use Hölder's inequality with the exponents $\frac{p}{q}$ and $\frac{p}{p-q}$.

Exercise 1.9.7 (Weak derivative and H^1 -regularity of $|x|$). Let $\Omega = (-1, 1) \subset \mathbb{R}$ and consider the function

$$u(x) = |x|.$$

1. Show that $u \in L^2(\Omega)$.
2. Define a candidate function $v(x)$ by

$$v(x) := \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$

Show that $v \in L^2(\Omega)$.

3. Prove that v is the weak derivative of u by checking that

$$\int_{\Omega} u(x) \varphi'(x) dx = - \int_{\Omega} v(x) \varphi(x) dx$$

for all test functions $\varphi \in C_c^\infty(\Omega)$.

4. Conclude that $u \in H^1(\Omega)$.

Hint. Use integration by parts on the intervals $(-1, 0)$ and $(0, 1)$ and note that φ vanishes at the endpoints.

Exercise 1.9.8 (Weak derivative and H^1 -regularity of $\mathbf{u}(\mathbf{x}) = \mathbf{x}|\mathbf{x}|$). Let $\Omega = (-1, 1)$ and consider

$$u(x) = x|x|.$$

1. Show that $u \in L^2(\Omega)$.
2. Define a candidate weak derivative

$$v(x) = \frac{d}{dx}(x|x|) = \begin{cases} 2x, & x > 0, \\ -2x, & x < 0. \end{cases} = 2|x|$$

Check that $v \in L^2(\Omega)$.

3. Verify that v is the weak derivative of u , i.e.

$$\int_{\Omega} u(x) \varphi'(x) dx = - \int_{\Omega} v(x) \varphi(x) dx \quad \forall \varphi \in C_c^\infty(\Omega).$$

4. Conclude that $u \in H^1(\Omega)$.

Exercise 1.9.9 (Weak derivative and H^1 -regularity of $\mathbf{u}(\mathbf{x}) = \sqrt{|\mathbf{x}|}$). Let $\Omega = (-1, 1)$ and consider

$$u(x) = \sqrt{|x|}.$$

1. Show that $u \in L^2(\Omega)$.
2. Try to define a candidate weak derivative

$$v(x) = \frac{d}{dx} \sqrt{|x|} = \begin{cases} \frac{1}{2\sqrt{x}}, & x > 0, \\ -\frac{1}{2\sqrt{-x}}, & x < 0. \end{cases}$$

Determine whether $v \in L^2(\Omega)$.

3. Conclude whether $u \in H^1(\Omega)$ or not, and explain why.

Exercise 1.9.10 (Weak derivative and H^1 -regularity for a function with singularity). Let $\Omega = (-1, 1) \times (-1, 1)^{N-1} \subset \mathbb{R}^N$, and write $x = (x_1, x')$ with $x_1 \in (-1, 1)$ and $x' \in (-1, 1)^{N-1}$. Consider the function

$$u(x) = |x'|^\alpha, \quad \alpha > 0,$$

which is singular along the line $x' = 0$.

1. Show that $u \in L^2(\Omega)$ if and only if $\alpha > \dots$

2. Compute formally the partial derivatives with respect to x' :

$$\partial_{x'_i} u(x) = \alpha |x'|^{\alpha-2} x'_i, \quad i = 1, \dots, N-1.$$

3. Determine for which values of α these derivatives belong to $L^2(\Omega)$.

4. Conclude for which α the function u belongs to $H^1(\Omega)$.

Hint. Use polar coordinates in $\mathbb{R}^{n-1}$ for the x' variables and integrate $|x'|^{2\alpha}$ and $|x'|^{2(\alpha-1)}$ over the unit ball.

Exercise 1.9.11 (Sobolev function with singularity in dimension 2). Let $\Omega = B(0, 1) \subset \mathbb{R}^2$ and consider the function

$$u(x) = \log\left(\frac{\log |x|^2}{2}\right).$$

1. Show that $u \in L^2(\Omega)$. (Hint: use polar coordinates $x = r(\cos \theta, \sin \theta)$ and check the integrability near $r = 0$.)

2. Compute the gradient of u formally for $x \neq 0$:

$$\nabla u(x) = \frac{1}{\log(|x|^2/2)} \frac{2x}{|x|^2}.$$

3. Show that $\nabla u \in L^2(\Omega)$. (Hint: integrate in polar coordinates and analyze the singularity at $r = 0$.)

4. Conclude that $u \in H^1(\Omega)$ despite being not continuous at $x = 0$.

Remark. This example illustrates that in dimension $N = 2$, functions in H^1 need not be continuous neither bounded, unlike in dimension 1.

Exercise 1.9.12 (Hilbert space structure from a symmetric bilinear form). Let H be a vector space and let

$$B : H \times H \rightarrow \mathbb{R}$$

be a bilinear form that is ***symmetric*** and ***coercive***, i.e.,

$$B(u, v) = B(v, u), \quad B(u, u) \geq \beta \|u\|^2$$

for all $u, v \in H$ and some $\beta > 0$.

1. Show that the map

$$(u, v) \mapsto B(u, v)$$

defines an ***inner product*** on H .

2. Deduce that H , equipped with the norm $\|u\|_B := \sqrt{B(u, u)}$, is a Hilbert space.

3. Give an example of a finite-dimensional $H = \mathbb{R}^N$ and a symmetric, coercive bilinear form B , and compute the corresponding norm explicitly.

Hint. Recall that an inner product must be symmetric, bilinear, and positive definite. Use the coercivity of B to show positive definiteness.

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